Chapter 5 – Statistical Inference and Analysis

Estimation, Tests and Regression with Python

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Learning Objectives

- Distinguish a population parameter from a statistic computed on a sample.
- Estimate means, proportions and variances, and quantify uncertainty.
- Formulate tests and interpret errors, power and p-values.
- Fit and assess simple and multiple linear regression models.
- Implement key procedures in Python.

Outline I

1. Sampling and Statistical Inference
2. Statistics and Sampling Distributions
3. Properties of Estimators
4. Maximum Likelihood Estimation
5. Confidence Intervals
6. Hypothesis Testing
7. Parametric Tests
8. Non-Parametric Tests
9. Simple Linear Regression
10. Model Evaluation Metrics
11. Multiple Linear Regression
12. Conclusion

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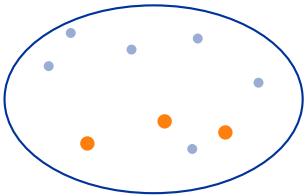
From a Sample to a Population

Definition

A **population** Ω contains all individuals under study. A **sample** $S = (\omega_1, \ldots, \omega_n)$ contains n selected individuals.

Factory example

Measure 50 rods to estimate the average length of all rods produced.



population Ω
orange: sample

Random Sample and Parameter

Definition

$X_1, \ldots, X_n$ form a random sample when they are independent and identically distributed (i.i.d.) as the population variable X.

- A **parameter**, such as $\mu = \mathbb{E}[X]$, describes the population.
- A **statistic**, such as $\bar{X}$, is computed from the random sample.
- The observed $x_1, \ldots, x_n$ are realizations of $X_1, \ldots, X_n$.

Inference uses a statistic to estimate a parameter or test a claim about it.

Two Tasks of Inference

Estimation

What is the population mean lifetime μ?
Report a point estimate and an interval.

Hypothesis testing

Is the claimed mean lifetime of 1000 hours compatible with the data?

Sample $\longrightarrow$ statistic $\longrightarrow$ inference about the population

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Sample Mean and Its Distribution

Definition

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

For i.i.d. observations with mean μ and variance σ^2,

$$\mathbb{E}[\bar{X}] = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Larger samples make the sample mean less variable.

Normality and the Central Limit Theorem

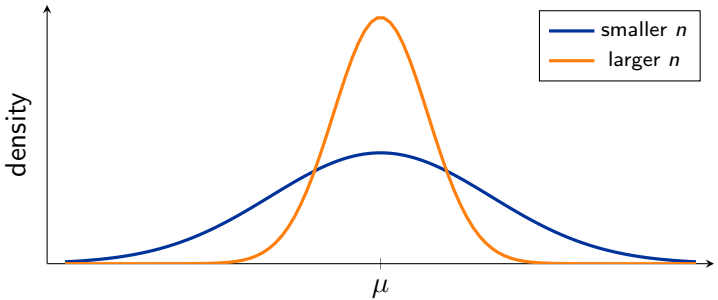
- If $X_i \sim N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \sigma^2/n)$ exactly.
- With finite variance, the central limit theorem gives an approximate normal distribution for $\bar{X}$ as n increases.

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

Battery lifetime

If $\sigma = 120$ hours and $n = 36$, the standard error is $120/\sqrt{36} = 20$ hours.

Sampling Distributions: Visual Intuition



Both distributions are centered at μ; the larger sample has smaller spread.

Sample Variance

Two denominators

$$s_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

- s_n^2 is the empirical variance; its expectation is $\frac{n-1}{n}\sigma^2$.
- S^2 is the corrected sample variance; $\mathbb{E}[S^2] = \sigma^2$.
- Under normal sampling, $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$.

Estimator and Estimate

Definition

An **estimator** $\widehat{\theta} = T(X_1, \ldots, X_n)$ is a statistic used for an unknown parameter θ. Its realized numerical value is an **estimate**.

Parameter	Estimator	Example estimate
μ	$\bar{X}$	$\bar{x} = 13.2$
σ^2	S^2	$s^2 = 4.7$
p	$\widehat{p} = \frac{1}{n}\sum_i X_i$	$\widehat{p} = .42$

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Bias and Mean Squared Error

$$\text{Bias}(\widehat{\theta}) = \mathbb{E}[\widehat{\theta}] - \theta.$$

Definition

An estimator is **unbiased** when its bias is zero.

$$\text{MSE}(\widehat{\theta}) = \mathbb{E}[(\widehat{\theta} - \theta)^2] = \text{Var}(\widehat{\theta}) + \text{Bias}(\widehat{\theta})^2.$$

MSE measures both random variation and systematic error.

Why the Variance Uses $n-1$

$$\sum_{i=1}^{n}(X_i-\bar{X})^2=\sum_{i=1}^{n}(X_i-\mu)^2-n(\bar{X}-\mu)^2.$$

Taking expectations gives $\mathbb{E}[\sum_i(X_i-\bar{X})^2]=(n-1)\sigma^2$.

Correction

$$\mathbb{E}\left[\frac{1}{n}\sum_i(X_i-\bar{X})^2\right]=\frac{n-1}{n}\sigma^2, \qquad \mathbb{E}[S^2]=\sigma^2.$$

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Likelihood Function

Given a parametric model $f(x;\theta)$ and independent observations,

$$L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i;\theta), \qquad \ell(\theta) = \log L(\theta).$$

Definition

The maximum likelihood estimator (MLE) maximizes L or, equivalently, ℓ.

A large likelihood means the observed sample is relatively plausible under the chosen parameter.

MLE: Bernoulli Example

For $X_i \sim \text{Bernoulli}(p)$, with $k = \sum_i x_i$ successes,

$$L(p) = p^k(1-p)^{n-k}, \qquad \ell(p) = k \log p + (n-k)\log(1-p).$$

For an interior optimum, $\ell'(p) = k/p - (n-k)/(1-p) = 0$, hence

$$\boxed{\widehat{p}_{\text{ML}} = \frac{k}{n} = \bar{x}}.$$

The boundary cases $k = 0$ and $k = n$ give $\widehat{p} = 0$ and 1.

MLE: Normal Mean and Variance

For i.i.d. $N(\mu, \sigma^2)$ observations,

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i (x_i - \mu)^2.$$

$$\widehat{\mu}_{\mathrm{ML}} = \bar{x}, \qquad \widehat{\sigma^2}_{\mathrm{ML}} = \frac{1}{n}\sum_i (x_i - \bar{x})^2.$$

The MLE for σ^2 uses n; the unbiased estimator uses $n - 1$.

Maximum Likelihood in Python

```
import numpy as np
x = np.array([2.1, 1.7, 2.4, 2.0, 1.8])
mu_mle = x.mean()
var_mle = x.var(ddof=0)
var_unbiased = x.var(ddof=1)
print(mu_mle, var_mle, var_unbiased)
```

Check the denominator when comparing software output to a formula.

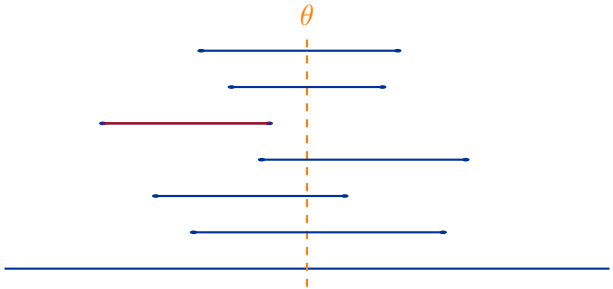
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What a 95% Confidence Interval Means

Definition

An interval procedure has 95% coverage if, over repeated random samples, 95% of its computed intervals contain the fixed parameter.



Mean: Known Variance

If X_i are normal and σ is known, the $(1-\alpha)$ interval is

$$\boxed{\bar{x} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}}.$$

The same formula is approximate for sufficiently large n under the CLT.

Calculation

$\bar{x} = 100$, $\sigma = 12$, $n = 36$: $100 \pm 1.96(12/6) = [96.08, 103.92]$.

Mean: Unknown Variance

For a normal population with unknown σ,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}, \qquad \boxed{\bar{x} \pm t_{n-1,1-\alpha/2} \frac{s}{\sqrt{n}}}.$$

Estimate the standard error with $s/\sqrt{n}$ and use $n-1$ degrees of freedom.

Mean Interval: Worked Example

A sample of $n = 25$ measurements has $\bar{x} = 52$ and $s = 10$. For 95% confidence, $t_{24,.975} \approx 2.064$.

$$\bar{x} \pm t_{24,.975} \frac{s}{\sqrt{n}} = 52 \pm 2.064 \frac{10}{5} = 52 \pm 4.128.$$

Interval

$$[47.87, 56.13].$$

Proportion Interval

For binary observations, $\widehat{p} = k/n$. When the normal approximation is adequate, the Wald interval is

$$\boxed{\widehat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}}.$$

Survey

$k = 120$ of $n = 200$: $\widehat{p} = .60$; the 95% margin is $1.96\sqrt{.6(.4)/200} \approx .068$. Interval $[.532, .668]$.

Near 0 or 1, or with small n, prefer a Wilson interval.

Variance Interval

For a **normal population**, $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. Writing $q_u = \chi^2_{n-1,u}$ for the lower-tail u quantile,

$$\left[\frac{(n-1)s^2}{q_{1-\alpha/2}},\ \frac{(n-1)s^2}{q_{\alpha/2}}\right].$$

The chi-square quantiles appear in reverse order because σ^2 is in the denominator of the pivot.

Confidence Intervals in Python

```
import numpy as np
from scipy import stats
x = np.array([49, 51, 55, 47, 58, 52, 51, 53])
n = len(x)
mean, s = x.mean(), x.std(ddof=1)
ci = stats.t.interval(.95, df=n-1,
                      loc=mean, scale=s / np.sqrt(n))
print(ci)
```

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Null and Alternative Hypotheses

Definition

H_0 describes a reference claim about the population; H_1 describes the alternative being tested.

Server response time

$H_0 : \mu = 200\,\mathrm{ms}$ versus $H_1 : \mu > 200\,\mathrm{ms}$.

The test compares observed evidence with what H_0 predicts. Failure to reject H_0 does not establish that it is true.

Two Types of Error

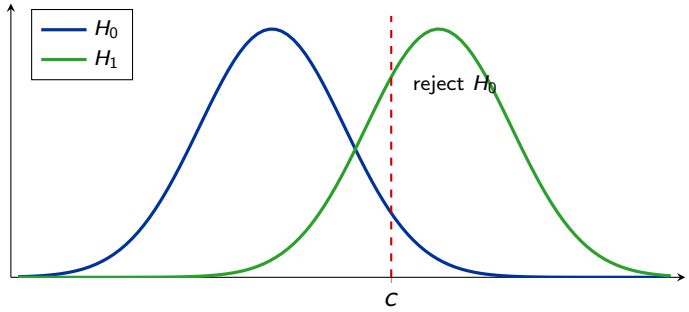
	H_0 true	H_0 false
Reject H_0	Type I error (α)	Correct rejection (power)
Do not reject H_0	Correct decision	Type II error (β)

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}), \qquad \beta = P(\text{do not reject } H_0 \mid H_1 \text{ true}).$$

Power

$$\text{Power} = 1 - \beta.$$

Decision Regions and Power



Under H_0, area right of c is α; under H_1, it is power.

A Test in Five Steps

1. State H_0, H_1 and the significance level α.
2. Select a statistic and its distribution under H_0.
3. Compute the statistic from the sample.
4. Calculate the p-value or compare with the critical region.
5. State a decision and interpret it in context.

p-Value

Definition

The p-value is the probability, *assuming* H_0, of results at least as extreme as the one observed, in the direction specified by H_1.

Decision rule

Reject H_0 when $p \leq \alpha$.

A p-value is not the probability that H_0 is true.

One-Sided and Two-Sided Tests

Alternative	Statistic z_{obs}	p-value under H_0
$\mu > \mu_0$	positive evidence	$1 - \Phi(z_{\text{obs}})$
$\mu < \mu_0$	negative evidence	$\Phi(z_{\text{obs}})$
$\mu \neq \mu_0$	either direction	$2[1 - \Phi(\lvert z_{\text{obs}} \rvert)]$

Choose the alternative before looking at the data.

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Z-Test for a Mean

Assume σ is known and observations are normal (or a large-sample approximation applies).

$$H_0 : \mu = \mu_0, \qquad Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1) \quad \text{under } H_0.$$

For $H_1 : \mu \neq \mu_0$, reject at level α if $|z_{\text{obs}}| > z_{1-\alpha/2}$.

Z-Test: Worked Example

$$H_0 : \mu = 100, \quad H_1 : \mu > 100, \quad \sigma = 12,\ n = 36,\ \bar{x} = 104.$$

$$z_{\text{obs}} = \frac{104 - 100}{12/6} = 2.$$

At level $\alpha = .05$, $z_{.95} = 1.645$. Since $2 > 1.645$, reject H_0.

Interpretation

The sample provides evidence that the population mean exceeds 100.

Student t-Test for a Mean

When σ is unknown, use

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \quad \text{under } H_0 \text{ for normal observations.}$$

For $H_1 : \mu \neq \mu_0$, reject if $|t_{\text{obs}}| > t_{n-1,1-\alpha/2}$.

For nonnormal observations, large-sample inference is approximate.

Which Test?

Situation	Statistic	Reference law
Known σ	$(\bar{X} - \mu_0)/(\sigma/\sqrt{n})$	standard normal
Unknown σ, normal sample	$(\bar{X} - \mu_0)/(S/\sqrt{n})$	t_{n-1}
Normal sample, variance claim	$(n-1)S^2/\sigma_0^2$	χ^2_{n-1}

Chi-Square Test for a Variance

Under normal sampling and $H_0 : \sigma^2 = \sigma_0^2$,

$$Q = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_{n-1}.$$

For a two-sided alternative, reject in either tail:

$$Q < \chi^2_{n-1,\alpha/2} \quad \text{or} \quad Q > \chi^2_{n-1,1-\alpha/2}.$$

This test is sensitive to departures from normality.

Parametric Tests in Python

```
from scipy import stats
import numpy as np
x = np.array([51, 55, 49, 54, 52, 50, 56, 53])
# Two-sided one-sample Student test: H0: mu = 50
result = stats.ttest_1samp(x, popmean=50)
print(result.statistic, result.pvalue)
```

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Sign Test

For independent observations and a continuous distribution, test a median m_0 by counting values above it.

$$B = \sum_{i=1}^{n} \mathbb{1}\{X_i > m_0\}.$$

After excluding ties, $B \sim \mathrm{Binomial}(n, 1/2)$ under H_0.

The sign test uses directions; it does not use the sizes of deviations.

Mann–Whitney Test

- Compare two independent samples by pooling and ranking observations.
- The rank sum measures whether one group tends to have larger values.
- A distribution-free comparison requires exchangeability under H_0; interpreting it specifically as a median test needs additional shape assumptions.

Application

Compare response times of two algorithms when extreme observations make a normal model questionable.

Non-Parametric Tests in Python

```
from scipy import stats
a = [13, 11, 15, 12, 14]
b = [17, 16, 19, 15, 18]
result = stats.mannwhitneyu(a, b,
                            alternative="two-sided")
print(result.statistic, result.pvalue)
```

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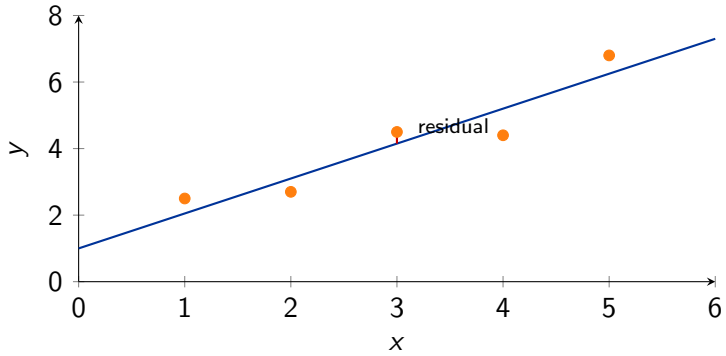
Definition

Simple linear regression models a response Y_i using one predictor x_i:

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

- β_0: expected response at $x = 0$.
- β_1: change in the expected response for a one-unit increase in x.
- ε_i: deviation from the regression line.

Least Squares: Fit a Line



Choose $\widehat{\beta}_0, \widehat{\beta}_1$ to minimize $\sum_i (y_i - \widehat{y}_i)^2$.

OLS Formulas

$$\widehat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}.$$

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i, \qquad e_i = y_i - \widehat{y}_i.$$

The slope is defined when the predictor values are not all equal.

Deriving the Slope

Minimize $Q(b_0, b_1) = \sum_i (y_i - b_0 - b_1 x_i)^2$.

$$\frac{\partial Q}{\partial b_0} = 0 \;\Rightarrow\; b_0 = \bar{y} - b_1 \bar{x}.$$

Substitute into $\partial Q / \partial b_1 = 0$:

$$b_1 \sum_i (x_i - \bar{x})^2 = \sum_i (x_i - \bar{x})(y_i - \bar{y}).$$

Solution

$b_1 = \widehat{\beta}_1$ and $b_0 = \widehat{\beta}_0$.

A Numerical Regression Example

x_i	1	2	3	4
y_i	2	3	5	4

$\bar{x} = 2.5$, $\bar{y} = 3.5$, $\sum(x_i - \bar{x})^2 = 5$, and $\sum(x_i - \bar{x})(y_i - \bar{y}) = 4$.

$$\widehat{\beta}_1 = 4/5 = .8, \qquad \widehat{\beta}_0 = 3.5 - .8(2.5) = 1.5.$$

$$\boxed{\widehat{y} = 1.5 + .8x}.$$

Decomposing Variation

With an intercept,

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum_i (\widehat{y}_i - \bar{y})^2}_{\text{SSR}} + \underbrace{\sum_i (y_i - \widehat{y}_i)^2}_{\text{SSE}}.$$

$$\boxed{R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = \frac{\text{SSR}}{\text{SST}}}.$$

R^2 is the fraction of in-sample variation in y explained by the fitted line.

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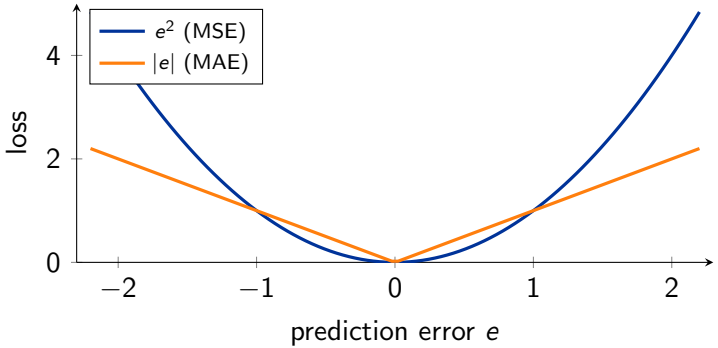
MSE, RMSE and MAE

For evaluation observations y_i and predictions $\widehat{y}_i$,

$$\mathrm{MSE} = \frac{1}{n}\sum_i (y_i - \widehat{y}_i)^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \qquad \mathrm{MAE} = \frac{1}{n}\sum_i |y_i - \widehat{y}_i|.$$

- MSE penalizes large errors quadratically.
- RMSE and MAE have the response variable's units.

Compare the Loss Functions



Squared error grows faster when a prediction is far from its target.

Evaluate on Unseen Data

- Calculate metrics on validation or test observations to estimate predictive performance.
- Training error is usually optimistic because the model was fitted to those observations.
- Compare models on the same test set and use the same target units.

Choose a metric that reflects the practical cost of prediction errors.

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Several Predictors

Definition

With p predictors,

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i.$$

β_j describes the change in the conditional mean for one unit of x_j *while holding the other included predictors fixed*.

House prices

Predict price using floor area, age and distance to the city center.

Matrix Form and OLS

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad X \in \mathbb{R}^{n\times(p+1)}, \qquad \boldsymbol{\beta} \in \mathbb{R}^{p+1}.$$

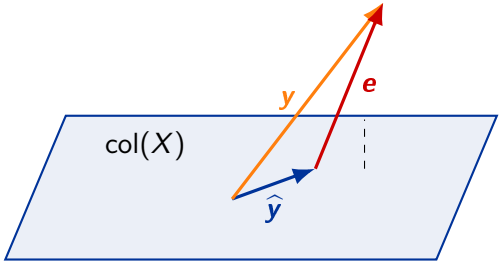
The first column of X consists of ones for the intercept. Minimize

$$\|\boldsymbol{y} - X\boldsymbol{b}\|_2^2.$$

If X has full column rank,

$$\boxed{\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \boldsymbol{y}}.$$

Projection Geometry



$\widehat{\boldsymbol{y}} = X\widehat{\boldsymbol{\beta}}$ is the projection of $\boldsymbol{y}$ onto the column space of X; $X^\top \boldsymbol{e} = 0$.

Regression Assumptions and Limits

- Linearity concerns the coefficients β_j, with predictors treated as specified in the model.
- $\mathbb{E}[\varepsilon \mid X] = 0$ supports unbiased coefficient estimation.
- Correlated predictors can make individual coefficients unstable.
- Error normality is useful for exact small-sample t inference, not required to calculate OLS coefficients.

Inspect residual patterns and validate prediction quality on new data.

Multiple Regression in Python

```
import statsmodels.api as sm
# df contains columns y, area, age, distance
X = sm.add_constant(df[["area", "age", "distance"]])
fit = sm.OLS(df["y"], X).fit()
print(fit.summary())
print(fit.predict(X.head()))
```

Use the coefficient table for inference and held-out data for prediction assessment.

From Inference to Machine Learning

Statistical inference

Estimate parameters, report uncertainty and test assumptions or claims.

Predictive modeling

Assess how well the model generalizes to unseen data.

A useful regression workflow connects both: fit, diagnose, quantify uncertainty and validate.

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Chapter 5: Key Formulas

$$\bar{X} = n^{-1} \sum_i X_i, \qquad S^2 = (n-1)^{-1} \sum_i (X_i - \bar{X})^2,$$

$$\mathrm{MSE}(\widehat{\theta}) = \mathrm{Var}(\widehat{\theta}) + \mathrm{Bias}(\widehat{\theta})^2, \qquad T = (\bar{X} - \mu_0)/(S/\sqrt{n}),$$

$$\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \boldsymbol{y}, \qquad R^2 = 1 - \mathrm{SSE}/\mathrm{SST}.$$

Match each procedure to its assumptions, and interpret results in the language of the application.

Thank You!

Questions?

